Curried Functional Spaces

Roman Maksimovich

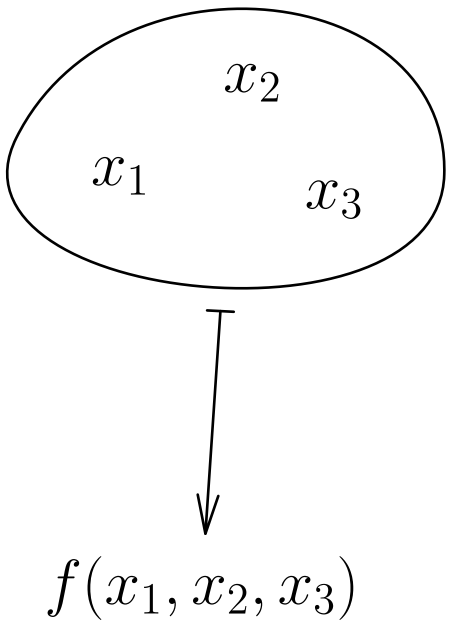
Hong Kong University of Science and Technology

May 9, 2026

Premise

Multivariable functions

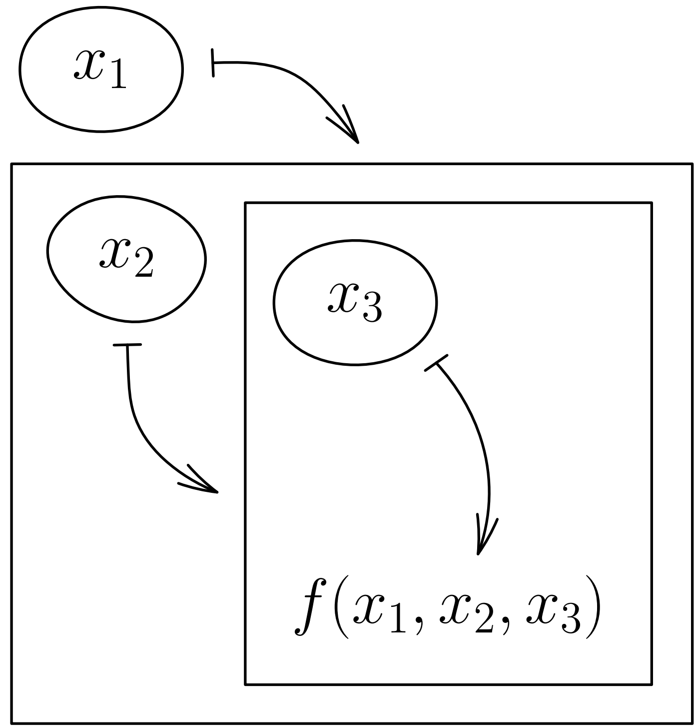
$\mathbb{R}^n \to \mathbb{R}$



Multivariate calculus

Curried functions

$\mathbb{R} \to (\mathbb{R} \to (\ldots(\mathbb{R} \to \mathbb{R})))$

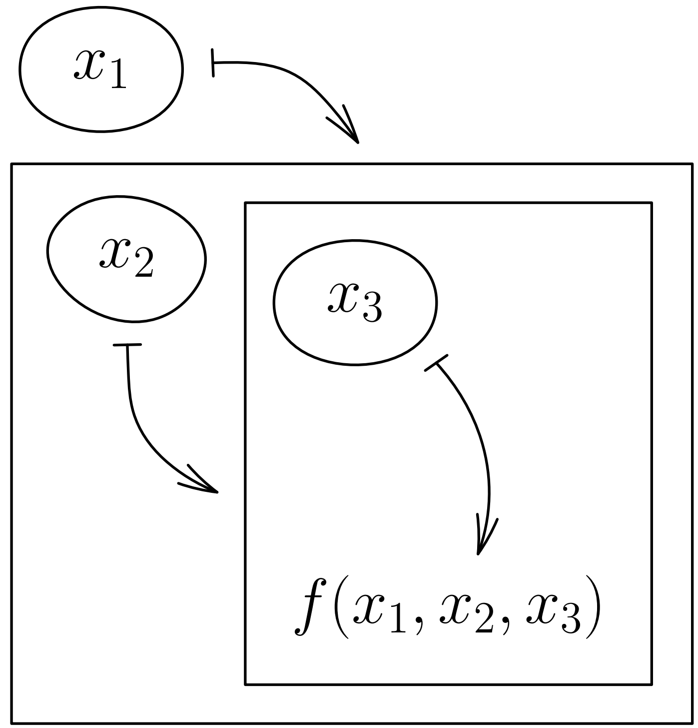


λ-calculus,
functional programming

Premise

Curried functions

$$\mathbb{R} \to (\mathbb{R} \to (\ldots(\mathbb{R} \to \mathbb{R})))$$



?

Multivariate calculus

λ-calculus,
functional programming

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n-ary curried functions

1. $C_0(\mathbb{R}) := \mathbb{R}$
2. $C_{n+1}(\mathbb{R}) := \{f : \mathbb{R} \to C_n(\mathbb{R}) \mid f \text{ is continuous}\}$

Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n-ary curried functions

1. $C_0(\mathbb{R}) := \mathbb{R}$

2. $C_{n+1}(\mathbb{R}) := \{f : \mathbb{R} \to C_n(\mathbb{R}) \mid f \text{ is } \underline{\text{continuous}}\}$

what is the topology on $C_n(\mathbb{R})$?

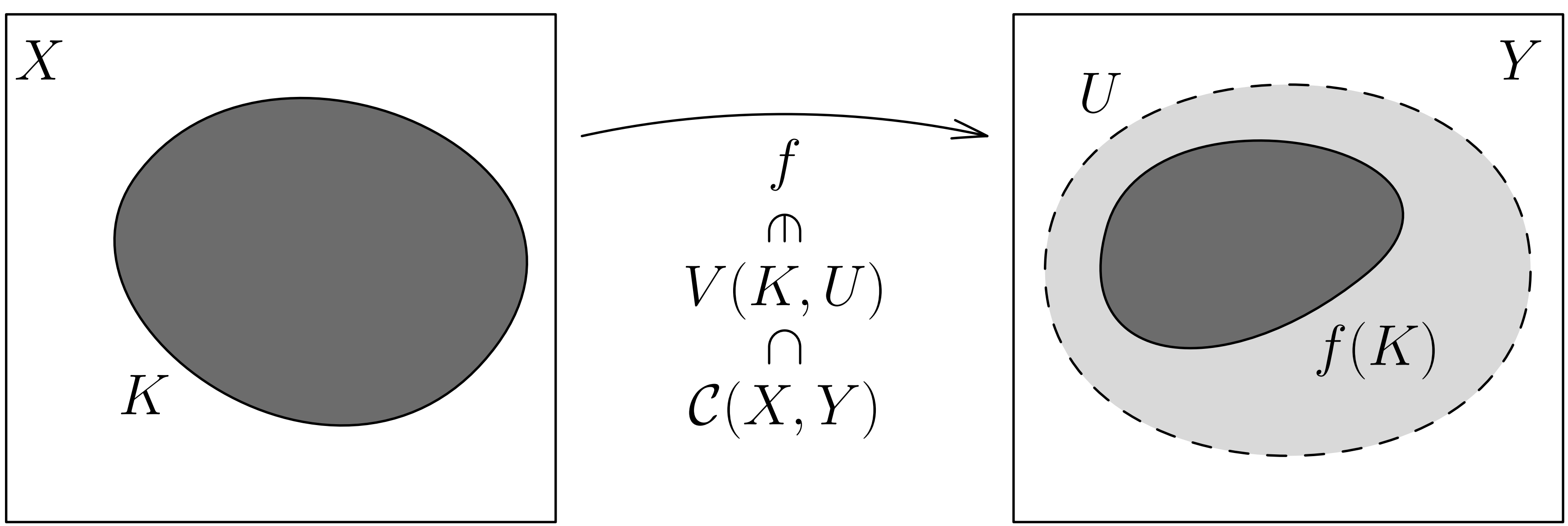
Continuous Functions

$C_n(\mathbb{R})$ — the space of continuous n-ary curried functions

1. $C_0(\mathbb{R}) := \mathbb{R}$

2. $C_{n+1}(\mathbb{R}) := \{f\colon \mathbb{R} \to \underline{C_n(\mathbb{R})} \mid f \text{ is continuous}\}$

compact-open topology: $C_n(\mathbb{R}) = \mathcal{C}(\mathbb{R}, C_{n-1}(\mathbb{R}))$

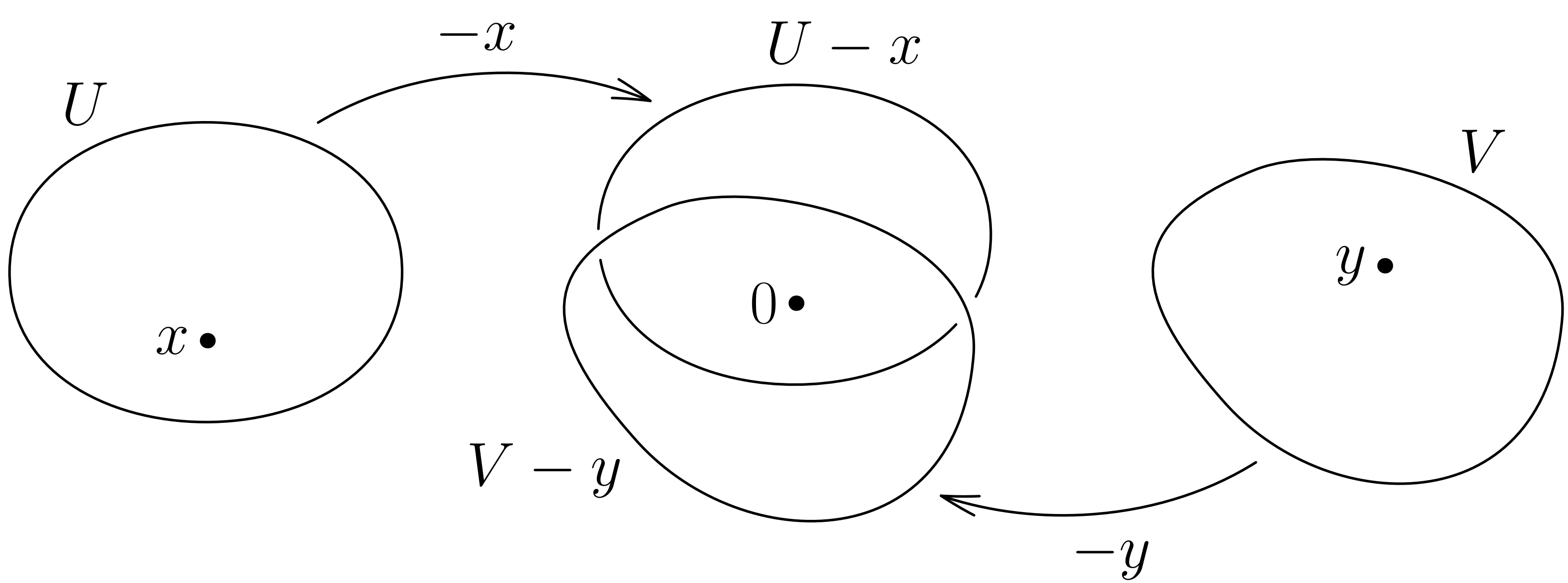


Topological Vector Spaces

Remark: If $f_1, f_2 \in C_n(\mathbb{R})$, then $\alpha f_1 + \beta f_2 \in C_n(\mathbb{R})$

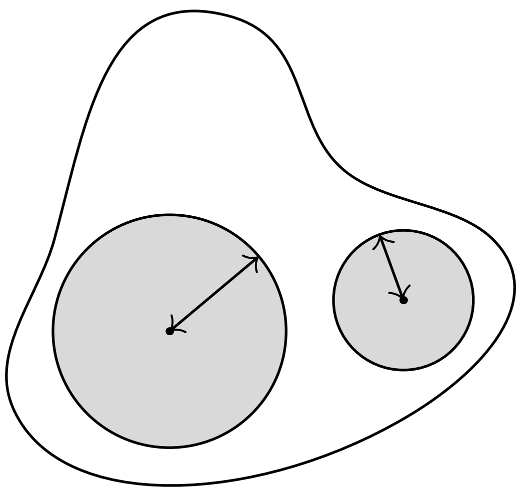
Definition: X is a TVS $\Leftrightarrow$ X has topological *and* vector structure, and they are *compatible*:

1. $(+)\colon X \times X \to X$ is continuous;
2. $(\cdot)\colon \mathbb{R} \times X \to X$ is continuous.



Properties of Topological Vector Spaces

Metrizability



Local convexity